

ARM-INTRODUCTION

KEY ASPECTS OF 8051

- **Core Architecture:** 8-bit CISC processor with a 16-bit program counter and 8-bit data bus.
- **Memory:** Typically features 4KB of on-chip ROM and 128 bytes of RAM, though this varies in later, enhanced versions.
- **Input/Output:** Generally includes four 8-bit I/O ports (32 pins).
- **Is features of 8051 sufficient for high performance embedded systems?**

How to choose your embedded processor?

- Depending on the end-use application, the processor is selected
- For example: When you need a robotic arm to pick and place a package from one place to another, then, ARM is not necessary, a simple PIC micro-controller will suffice
- Whereas when you want to deliver the package using drone to your home, then you choose ARM processor to direct the drone integrated with Google maps, detect obstacles and make rerouting when needed



Figure: Pick and place robot



Figure: Delivery drone

LIMITATIONS OF 8051:

- **Performance Bottleneck:** The 8051 is an 8-bit architecture. It struggles with tasks requiring high-speed data processing, complex calculations, or multi-tasking.
- **Memory Constraints:** The classic architecture supports only 4KB of on-chip ROM, which is too small for advanced firmware.
- **Power Consumption:** High on comparison with newer microcontrollers
- **Limited Peripherals:** Traditional 8051 lacks built-in modern interfaces like USB, Ethernet, or advanced Analog-to-Digital Converters (ADCs).
- **Development Limitations:** The 8051 C compiler often produces non-thread-safe code, making it unsuitable for applications requiring an RTOS (Real-Time Operating System)

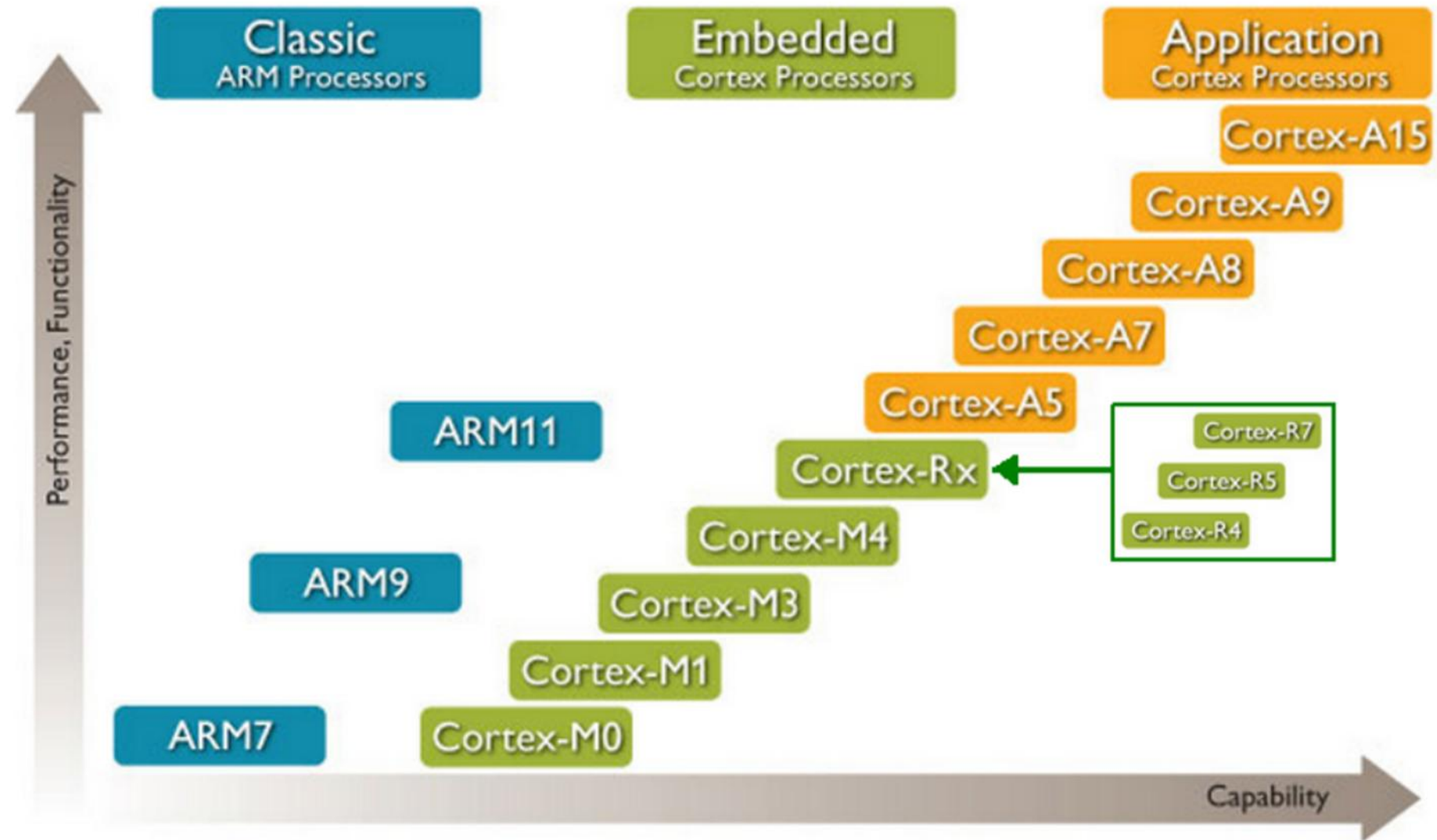
Introduction: Architecture of ARM microcontroller

- A microcontroller was developed by ACORN microcontrollers in the year 1983.
- The idea is to replace the 8-bit 6502 microprocessor in BBC computers.
- This led to the 1st RISC implementation.
- A company was founded in 1990 named ARM
- The name ARM is an acronym, expanded as **Advanced RISC Machine**.
- Initially this company was jointly owned by Acorn, Apple and VLSI.

Why ARM?

- ARM is a powerful **32-bit microcontroller with pipelining ability** that surpasses the previous 8-bit processor.
- They are mainly used in battery-operated devices since they are **less power consuming** processors.
- Some of the use cases of ARM include:
 - ARM 7: iPod
 - ARM 9: BenQ, Sony Ericsson
 - ARM 11: iPhone, Nokia N93, N100
- Upto 2010, 90% of the 32-bit embedded RISC processors are made of ARM.

Generations of ARM embedded processors



Evolution of ARM μ C

ARM 7

- 3-stage pipeline
- High code density, Low power density
- Used for low end systems (as it adapts Von Neumann Architecture)

ARM 9

- Compatible with ARM 7
- Cache memory came in
- Separate instruction cache and data cache were present
- Adapts Harvard Architecture

ARM 10

- In addition to ARM 7 features, it supports 6-stage pipelining
- The stages are: Fetch | Issue | Decode | Execute | Memory | Write

ARM processors' comparison

	ARM 7 (1995)	ARM 9 (1997)	ARM 10 (1999)	ARM 11 (2003)
Pipeline depth	3-stage	5-stage	6-stage	8-stage
Typical clock frequency (MHz)	80	150	260	335
Power (mW/MHz)	0.06	0.19	0.5	0.4
Throughput (MIPS/MHz)	0.97	1.1	1.3	1.2
Architecture	Von Neumann	Harvard	Harvard	Harvard
Multiplier	8 * 32	8 * 32	16 * 32	16 * 32

Features of ARM

- Energy efficiency: Less power consumption
- High Performance: video editing, Gaming
- Scalability and Customization
- **RISC architecture**
- Versatility: Applications in several sectors, including automotive, healthcare, aerospace, and consumer electronics.
- For example, ARM-based processors are used in the automotive industry to power advanced driver assistance systems (ADAS). They also play a critical role in the healthcare sector, where they are used in imaging equipment, prosthetics, and digital healthcare systems.

RISC Design Philosophy

RISC (Reduced Instruction Set Computer)

Simpler Instructions: RISC processors use a smaller set of simple instructions, making them easier to decode and execute quickly.

Single-Cycle Execution: Each instruction takes just one clock cycle to execute

Load/Store Architecture: Data access to memory is handled only through dedicated load and store instructions, with most operations performed on data held in registers.

More General-Purpose Registers: to minimize the memory access

RISC Design Philosophy

Simple Addressing Modes

Fewer Data Types: supports six data types namely 8 bit signed and unsigned, 16bit signed and unsigned plus 32 bit signed and unsigned.

Pipeline Efficiency

Hardware Simplicity/ Software Intelligence

RISC Design Philosophy

Advantages:

- Faster Execution.
- Lower Power Consumption.

Disadvantages:

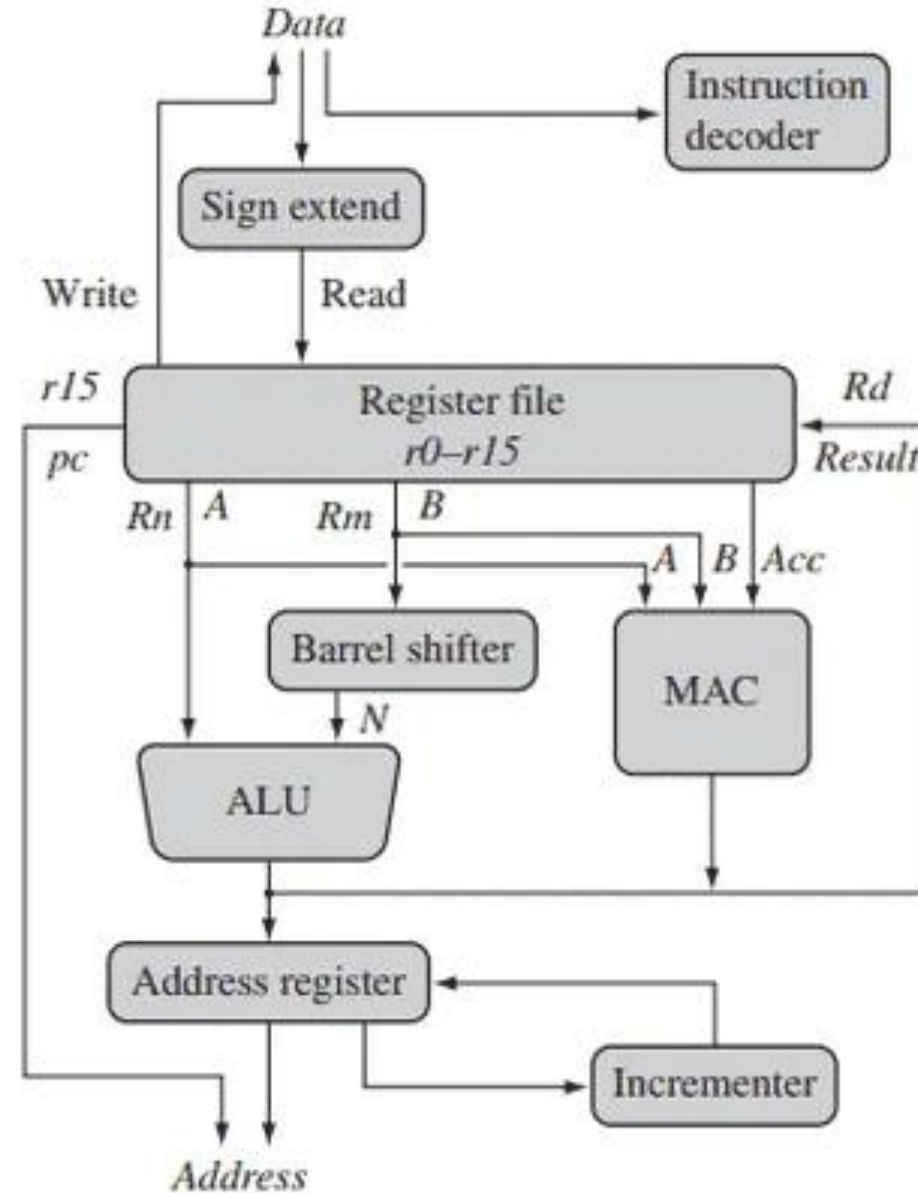
- More Instructions Required compared to CISC
- Increased Memory Usage : Additional instructions for complex tasks can lead to higher memory requirements.
- Higher Development Cost.

ARM7TDMI-S PROCESSOR

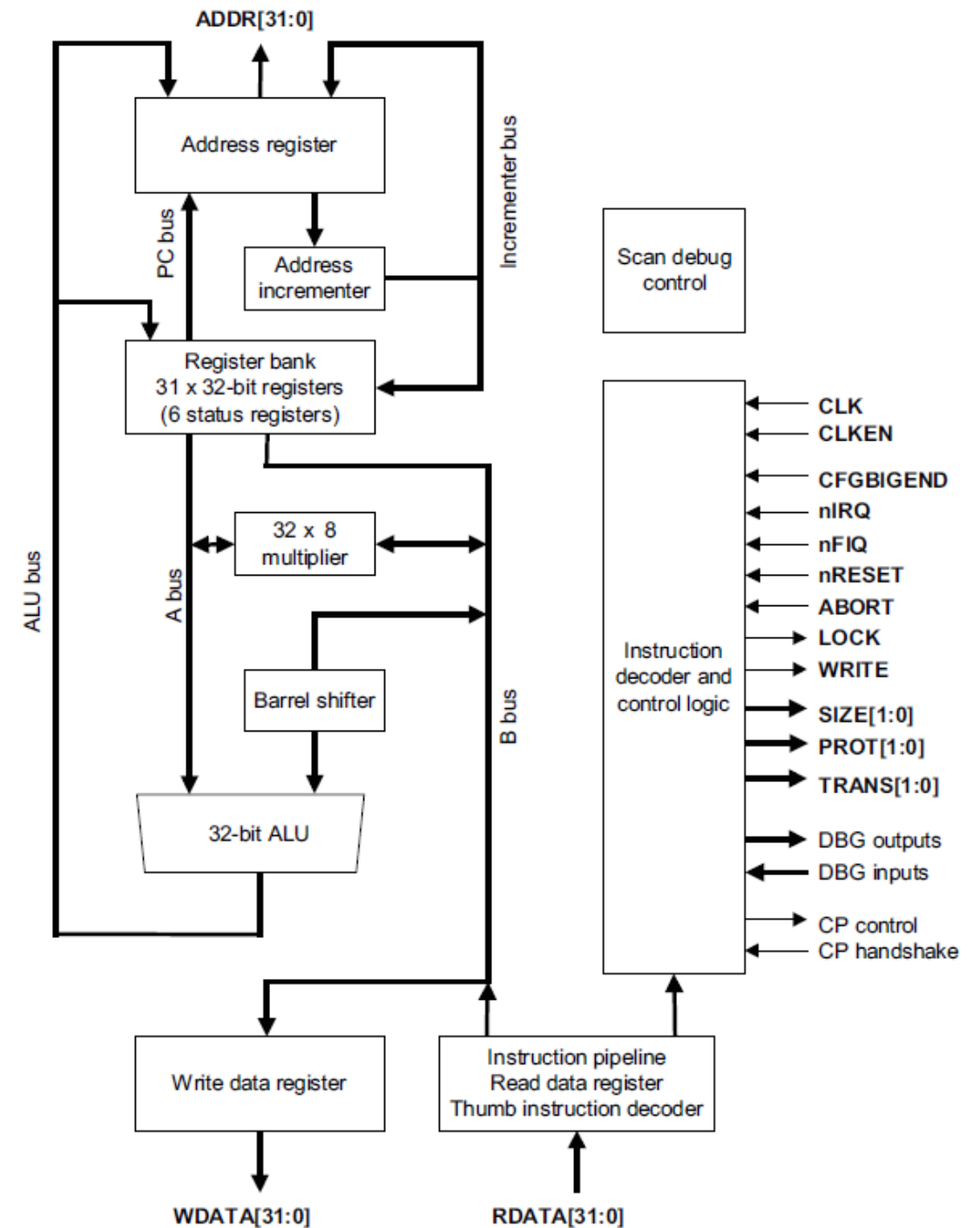
□ Key Components & Features (TDMI-S):

- ❖ **T (Thumb)**: Supports 16-bit instructions, improving code density by ~30% compared to standard 32-bit ARM instructions.
- ❖ **D (Debug)**: Includes embedded ICE (In-Circuit Emulation) logic for debugging M
- ❖ **M (Multiplier)**: Features an enhanced multiplier, which is usually a fast 32x8 hardware multiplier.
- ❖ **I (ICE)**: Supports on-chip debug capabilities (part of the D feature).

DATAFLOW MODEL OF ARM7 PROCESSOR



ARM7TDMI-S block diagram



Basic Processing Unit

ALU (Arithmetic and Logical Unit)

ALU is designed to conduct mathematical operations such as add, sub, etc., as well as logical operations such as OR/AND, etc., on data stored in data registers.

Barrel shifter

A barrel shifter is a digital circuit capable of shifting a data word by a specific number of bits in a single clock cycle. This hardware unit can execute data shifting operations on the data in the register bank.

Basic Processing Unit

MAC (Multiply and accumulate unit)

Basic summation operations on data in registers are supported.

Any operation's result can be written back to the register bank, or if the instruction needs memory access, the result is sent to the address register.

Address register

This contains the address from which data or instruction needs to be fetched.

This register is connected to an incrementing unit.

Internal Architecture of ARM7

- ❖ Von Neumann Architecture
- ❖ 3-stage pipeline
- ❖ 32-bit Data bus
- ❖ 32-bit Address bus
- ❖ 37 32-Bit Registers
- ❖ 32-bit ARM instruction Set
- ❖ 16-Bit Thumb instruction Set
- ❖ 32×8 Multiplier

ARM7TDMI-S PROCESSOR

- ❖ 32-bit microprocessors
- ❖ Reduced Instruction Set Computer (RISC)
- ❖ Von Neumann architecture, with a single 32-bit data bus carrying both instructions and data.
- ❖ A three-stage pipeline is used, so instructions are executed in three stages:
 - Fetch
 - Decode
 - Execute.
- ❖ Only load, store, and swap instructions can access data from memory.
- ❖ Data can be 8-bit bytes, 16-bit halfwords, or 32-bit words

Program Counter

- Note: Program Counter $PC = PC + 4$.
- A address incrementer circuit increments the address by 4 as word size of machine is 4 bytes.
- The memory in ARM processors are byte-addressable.

Programming model of ARM

ARM consists of special purpose registers and general purpose registers.

The special purpose registers are:

PC: Program counter [R15]

LR: Link register [R14]

- When a procedure call is made, the return address is automatically placed into this register. A return from the procedure can be implemented by copying R14 to R15

SP: Stack pointer [R13]

Programming model of ARM

CPSR: Current Program Status Register

SPSR: Saved Program Status Registers [5 in number]

- Holds a copy of the previous status register while executing exception or interrupt handler routines
- Copied back to CPSR on return from exception or interrupt

General Purpose Registers

- 6 datatypes are supported
- Byte operations (8-bit operations), half-word operations (16-bit operations) and word operations (32-bit operations) are supported
- All data transfer instructions use byte operations
- Whereas all arithmetic and logical operations use only word operations

CPSR of ARM processor

Current Program Status Register (CPSR) :: Flag Register

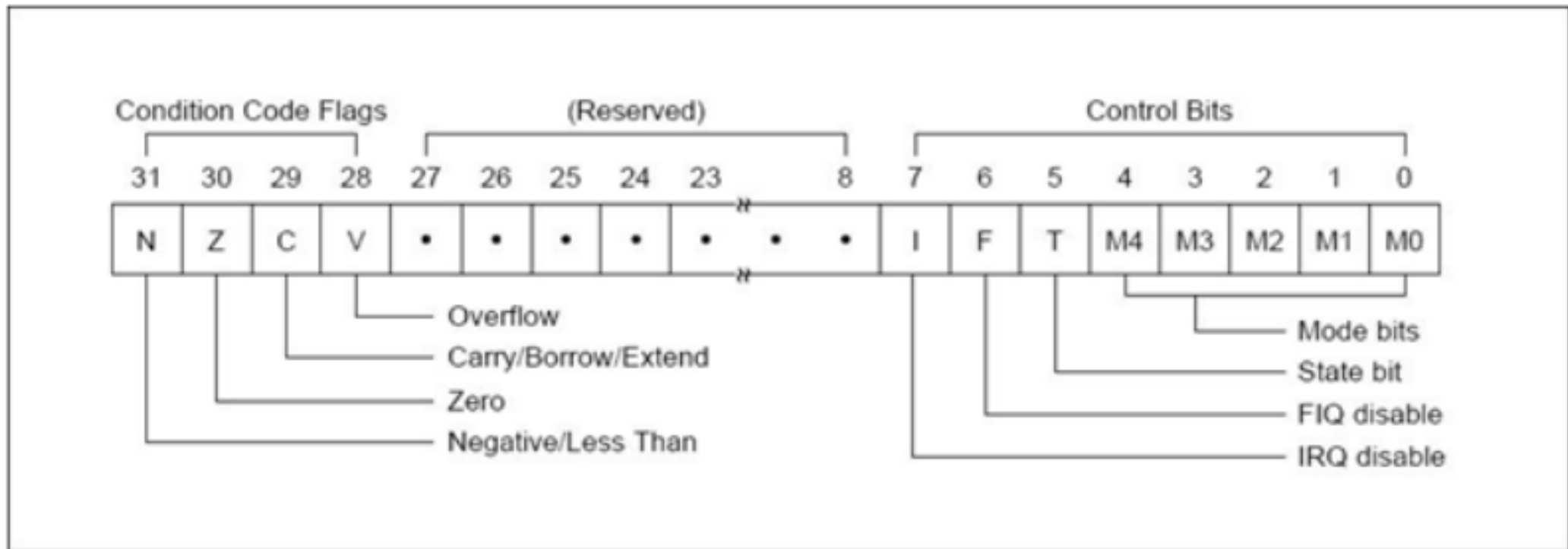
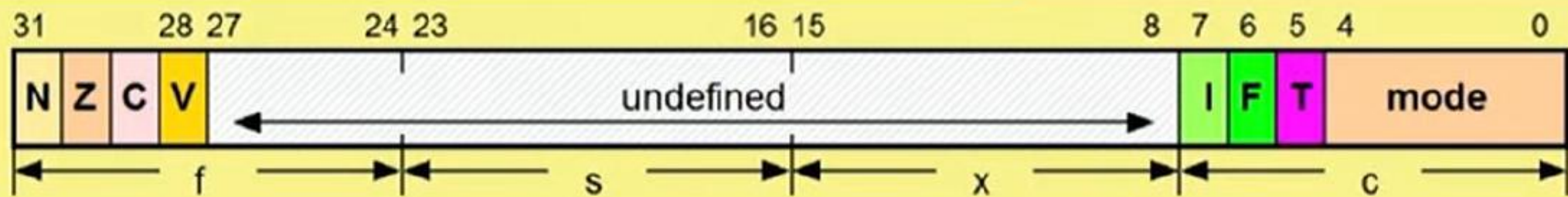


Figure: CPSR of ARM processor



Condition code flags

- N = **N**egative result from ALU
- Z = **Z**ero result from ALU
- C = ALU operation **C**arried out
- V = ALU operation **oV**erflowed

Mode bits

10000	User
10001	FIQ
10010	IRQ
10011	Supervisor
10111	Abort
11011	Undefined
11111	System

Interrupt Disable bits.

- I** = 1: Disables the IRQ.
- F** = 1: Disables the FIQ.

T Bit (Arch. with Thumb mode only)

- T = 0: Processor in ARM state
- T = 1: Processor in Thumb state

ARM processor modes

There are 7 processor modes for ARM. They are:

1. **USR: User mode**

- ❖ Default mode to run the application code
- ❖ CPSR cannot be written
- ❖ Mode can only be changed by exception generation

2. **Interrupts modes**

a. **FIQ: Fast Interrupt Request mode**

- ❖ Supports high speed interrupts handling. Generally used for a single critical interrupt source in a system.

b. **IRQ: Interrupt Request mode**

- ❖ For normal interrupts
- ❖ even maskable interrupts are enabled

3. **SVC: Supervisor mode**

- ❖ A protected mode for running system level code to run OS calls.
- ❖ processor enters this mode after a reset.

ARM processor modes

4. ABT: Abort mode

- ❖ During memory access violations, an abort exception will be generated

5. UND: Undefined mode

- ❖ If a fetched opcode is not an ARM instruction, an undefined instruction exception will be generated

6. System mode:

- ❖ Runs privileged operating system tasks
- ❖ This mode is used in some ARM architectures to provide unrestricted access to all system resources, often used for debugging or low-level tasks.

- ❖ Non user modes (called privileged modes) are entered to serve interrupts or exceptions

Memory mapped IO technique

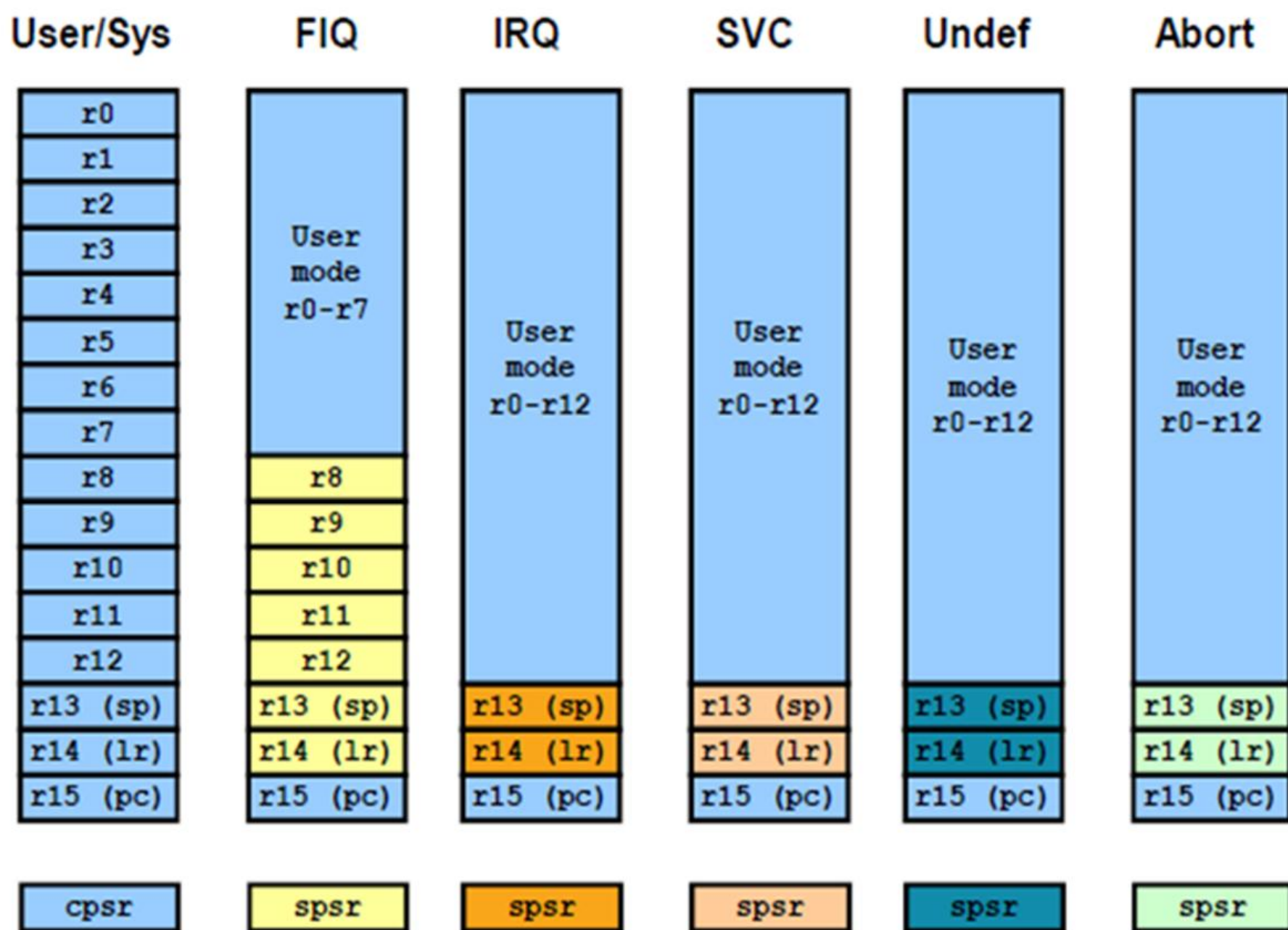
- Generally ARM processors connect with IO via memory mapped IO technique.
- So, only load and store instructions are used to access IO.
- Memory access violation: When a IO is referred by the address location of a memory, then it causes memory access violation, which makes the processor to go in "Abort" mode.

ARM REGISTERS

- Basically there are two types of registers – General purpose registers and Special purpose registers.
- The registers in are all 32 bits in size.
- In each mode, up to 18 active registers are available: 16 data registers and 2 processor status registers.
- The data registers are labeled r0 through r15 by the programmer.
- The ARM processor contains three registers: r13, r14, and r15, each of which is allocated to a specific duty or unique function.

Registers Overview:

- r0-r12 : General-purpose data storage.
- r13(SP): Stack Pointer, points to the top of the current stack.
- r14(LR): Link Register, stores the return address.
- r15(PC): Program Counter, holds the address of the next instruction.
- CPSR: Current Program Status Register, contains condition flags and mode bits.
- SPSR: Saved Program Status Register, stores the CPSR of the previous mode when an exception occur



Note: System mode uses the User mode register set

Exception handling

- When exception occurs,
 1. the CPSR is copied to SPSR
 2. Sets appropriate bits in CPSR
 - a) Changes to ARM state
 - b) Disables IRQ and FIQ
 3. Stores return address in LR
 4. Changes to related mode
- To return,
 - SPSR is copied back to CPSR
 - LR is copied to PC

ARM Execution modes

ARM mode

- All instructions must be 32-bit wide
- All instructions must be word aligned (Word aligned means all instructions must start from a memory address which is a multiple of 4)
- PC's last two bits are zeroes, to be a multiple of 4

Thumb mode

- Used in smaller applications
- All instructions are 16 bits wide
- All instructions are half-word aligned (instruction addresses are a multiple of 2)
- Last bit of PC is always zero

Comparison of ARM mode and Thumb mode

	ARM mode (CPSR T=0)	Thumb mode(CPSR T=1)
Instruction size	32-bit	16-bit
Core instructions	58	30
Conditional execution	Most	Only branch instructions
Data processing instructions	Access to barrel shifter and ALU	Separate barrel shifter and ALU instructions
Program status register	Read-write in privileged mode	No direct access
Register usage	15 GPRs + PC	8 GPRs + 7 high register + PC